Adria E. Brooks, Ph.D.
Power Systems & Renewable Energy

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Skilled Analyst with a broad range of experience in leading energy research and over 15 years of experience in the power systems and renewable energy sectors. Knowledgeable in technical project requirements including programming and engineering, with expertise in national power system models. Strong team leader and author of several different reports and peer-reviewed articles. Dedicated to supporting the evolution of the Nation's power grid. Skilled in effective written and oral communication, presenting technical material to diverse audiences, data visualization, programming in multiple languages, and bridging the divides between policy and engineering.

Areas of Expertise

- Electric Transmission Planning
- Power System Modeling
- Electricity Markets
- Clean Energy
- Energy Equity & Justice
- State & Federal Regulation
- Research & Analysis
- Report & Affidavit Authorship
- Client Relations
- Public Speaking
- Media Interviews
- Data Visualization

Professional Experience

Grid Strategies, Bethesda, Maryland **February 2025 – present**
Director of Transmission Planning

Advises clients on regional transmission planning processes, contributing to data analysis, stakeholder comments, and regulatory filings. Contributes to and authors reports on system planning best practices, transmission benefit evaluation, resource adequacy, and grid enhancing technologies. Maintains good relations with a broad client base, including utilities, infrastructure development, power system modelers, environmental organizations, and consumer advocates.

- Authored or contributed to four technical affidavits at the Federal Energy Regulatory Commission and seven technical reports on regional transmission planning.
- Consistently leads or contributes to more than one dozen projects at a time with different clients
- Has brought in and/or led contracts totaling $800,000 in revenue within first 18 months of employment.

U.S. Department of Energy, Washington, D.C. **October 2021 – February 2025**
Senior Technical Advisor in Transmission Planning

Key member of Grid Deployment Office with goal of enhancing the efficiency and reliability of the nation's power grid to meet decarbonization goals. Portfolio included national transmission planning given different policy and reliability constraints, analysis of power grid congestion and reliability concerns, management of several simultaneous research projects, technical stakeholder engagement, data center integration, and briefing leadership on grid needs.

- Led and advised research projects, including the *2023 National Transmission Needs Study*, which requires expert knowledge of the national power system modeling landscape and extensive stakeholder engagement. The *Needs Study* won "Best Published Research for a General Audience" from the American Energy Society.
- Led and participated in cross-cutting research teams on topics of interest to the White House and Secretary of Interest, including national transmission, data centers, and grid-enhancing technologies.
- Trusted subject matter expert and advisor to external transmission system studies, such as the NERC *Interregional Transfer Capability Study* and providing expert testimony at FERC technical conferences.
- Provided over one hundred briefings on technical transmission topics to the public, media, industry, and government leadership across the U.S. Briefings were frequently applauded by attendees as clear and engaging.

U.S. Department of Energy, Washington, D.C. **September 2020 – October 2021**
AAAS Science and Technology Policy Fellow

Provided analysis on top national renewable energy priorities, managed national lab research, and responded to Congressional or DOE leadership information requests. Projects included energy justice, electric transmission planning, electricity markets, and demand-side grid modeling.

- Facilitated more than a dozen workshops with 50+ attendees each, which involved developing workshop materials and coordinating schedules.
- Authored two reports for Congress within a few months and reviewed several dozen articles for leading power system journals.
- Managed a dozen projects simultaneously across different research groups.
- Developed a new energy justice roadmap project, which re-envisions a new suite of energy analysis tools focused on equity and justice.

Public Service Commission of Wisconsin, Madison, Wisconsin **January 2019 – August 2020**
Public Service Engineer

Spearheaded research projects and advised leadership regarding transmission planning and wholesale electricity market policies including issues with transmission planning, distributed energy resources, energy storage, and electric vehicles.

- Led multiple analyst and research projects.
- Advised Commissioners on transmission and renewable energy policy once per week.
- Wrote weekly internal policy memos to communicate transmission system material.

University of Wisconsin, Madison, Wisconsin **September 2015 – August 2020**
Graduate Research Fellow

Researched topics including electric power systems, electricity markets, optimal dispatch with variability, microgrids, renewable energy, and distributed energy resources. Oversaw a student project with the Midcontinent Independent System Operator (MISO) states to quantify the existence of distributed energy resources and to compare grid interconnection policies of other regional transmission organizations.

- Authored and published three peer reviewed papers and four conference papers.
- Led a weekly peer research group to produce a high quality report for a regulatory client.
- Coded for modeling research in four different programming languages.
- Reviewed several dozen articles for leading power system journals.

University of Arizona, Tucson, Arizona **May 2010 – September 2015**
Research Specialist

Led a team of eight students and post-graduate researchers. Installed and evaluated new photovoltaic (PV) systems throughout Southern Arizona, and analyzed system technologies for the benefit to electric utility companies, product developers, and the scientific community.

- Initiated 20+ different research projects to report performance and reliability measures of PV systems across Southern Arizona.
- Liaised with leaders of 12 utilities and solar energy companies to understand research needs.
- Authored four peer reviewed journal articles and 15 conference papers.
- Wrote three grant/contract applications awarding $395K to lab group.

Additional Experience

Research Fellow, Arizona Research Institute for Solar Energy, Tucson, Arizona
Engineering Intern, Semprius, Inc., Durham, North Carolina
Intern, National Renewable Energy Laboratory, Golden, Colorado
Environmental Conservation Intern, Student Conservation Association, AmeriCorps, Hawley, Massachusetts
Student Deputy Principal Investigator, Lunar and Planetary Laboratory, Tucson, Arizona
Research Fellow, National Space Science and Technology Center, Huntsville, Alabama

Education

Master of Science and Ph.D. in Electrical Engineering, August 2020
University of Wisconsin, Madison WI

Energy Analysis and Policy Certificate, December 2018
University of Wisconsin, Madison WI

Bachelor of Science in Engineering Physics, May 2011
University of Arizona, Tucson AZ

Publications

Textbook Chapters

[T1] **Brooks, A.E.,** "Solar Energy: Photovoltaics." *Future Energy, 2nd Ed*. Ed Trevor Letcher. New York: Elsevier, 2014. 383-404. Print.

Journal Articles

[J1] Mai, T., Brown, P.R., Palchak, D., Brinkman, G., **Brooks, A.E.,** Denholm, P., Gramlich, R. Homer, J.S., Lew, D., Kuna, J., Okullo, J., Pfeifenberger, J., Rose, A., Simeone, C., and Wright, J. "What can we learn from US national transmission studies?" Joule *10*, 102339 (Feb 2026)

[J2] Frew, B., Anwar, M.B., Dalvi, S., **Brooks, A**., "The interaction of wholesale electricity market structures under futures with decarbonization policy goals: A complexity conundrum." Applied Energy. Vol 339 (Mar 2023).

[J3] Du X., Zhao, Y., **Brooks, A.E**., "Wholesale Price Dynamics in the Evolving Texas Power Grid." *The Electricity Journal*, Vol. 36, No. 5 (Jun 2023).

[J4] Frew, B., Anwar, M.B., Dalvi, S., **Brooks, A.**, "The interaction of wholesale electricity market structures under futures with decarbonization policy goals: A complexity conundrum." *Applied Energy.* Vol 339 (Mar 2023).

[J5] **Brooks, A.E**., Lesieutre, B.C. "The validity of a locational marginal price on variable power injections in energy and regulation markets." *International Journal of Electrical Power & Energy Systems*. Vol 121 (Oct 2020).

[J6] Zielke, M., **Brooks, A.E**., Nemet, G. "The Impacts of Electrical Vehicle Growth on Wholesale Electricity Prices in Wisconsin." *World Electric Vehicle Journal*. Vol 11, No. 2 (Apr 2020).

[J7] **Brooks, A.E**., Lesieutre, B.C. "A Review of Frequency Regulation Markets in Three U.S. ISO/RTOs." *The Electricity Journal*, Vol. 32, No. 10 (Dec 2019).

[J8] Barron-Gafford, G.A., Minor, R.L., Allen N.A., Cronin, A.D., **Brooks, A.E**., Pavao-Zuckerman M.A. "The Photovoltaic Heat Island Effect: Large solar power plants increase local temperatures." *Scientific Reports,* 6, Article No. 35070 (Oct 2016).

[J9] Lonij, V.P.A **Brooks, A.E**., Cronin, A.D., Leuthold, M., Koch, K. "Intra-hour forecasts of solar power production using measurements from a network of irradiance sensors." *Solar Energy,* Vol. 97 (Nov 2013): 58-66.

[J10] **Brooks, A.E**., Allen, N., Lonij, V.P.A., Cronin, A.D. "Evaluation of Four Geomembrane-Mounted PV Systems for Land Reclamation in Southern Arizona." *Journal of Energy and Power Engineering,* Vol. 7, No. 5 (May 2013): 834-840.

Technical Reports & Affidavits

[R1] Ortiz, D.S. and **Brooks, A.E.** "A Case for Interregional Evaluation of the MISO-SPP Northern and Central Seam." Grid Strategies. (Oct 2026)

[R2] **Brooks, A.** "Testimony of Dr. Adria Brooks on Behalf of Interwest Energy Alliance." FERC Docket No. ER26-2801, ER26-2803, ER26-2804, ER26-2805, ER26-2806, ER26-2811, ER26-2816, ER26-2817, ER26-2822, ER26-2828, ER26-2831. (Aug 2026)

[R3] **Brooks, A.** "Testimony of Dr. Adria Brooks on Behalf of Interwest Energy Alliance." FERC Docket No. ER26-2793, ER26-2795, ER26-2802, ER26-2809, ER26-2812, ER26-2813, ER26-2818, ER26-2821. (Aug 2026)

[R4] **Brooks, A.** "Capacity Value of Interregional Transmission." Grid Strategies. (Jul 2026)

[R5] **Brooks, A.** "Review of NERC's 2026 Summer Reliability Assessment." Grid Strategies. (Jul 2026)

[R6] **Brooks, A.** "Modeling the Order 1920 Benefit Metrics." Grid Strategies. (May 2026)

[R7] **Brooks, A.** "Review of NERC's 2025 Long-Term Reliability Assessment." Grid Strategies. (Mar 2026)

[R8] Zimmerman, Z., **Brooks, A.,** Gramlich, R., Shreve, N., Jamal, D. "2025 Transmission Planning and Development Report Card." Grid Strategies. (Feb 2026)

[R9] **Brooks, A.,** Silverstein, A., Gramlich, R. "Resource Adequacy Value of Interregional Transmission." Grid Strategies. (Jun 2025)

[R10] Wiser, R., **Brooks, A.**, Donohoo-Vallet, P., Oskar, G., Frost, J., Brown, P., Mai, T. "Transmission impact assessment: Power sector infrastructure deployment to reduce costs, improve reliability, and lower pollution." U.S. Department of Energy (Oct 2024).

[R11] "National Transmission Needs Study." U.S. Department of Energy (Oct 2023). *[Served as Principal Investigator]*

[R12] **Brooks, A.E.** “Understanding Minimum Transfer Capability Requirements,” FERC Technical Conference, Docket No. AD23-3-000 (Dec 2022).

[R13] **Brooks, A.E.** “Renewable Energy Resource Assessment Information for the United States.” U.S. Department of Energy Office of Energy Efficiency and Renewable Energy (Mar 2022).

Conference Papers and Preprints

[C1] **Brooks, A.E.**, Lesieutre, B.C. “RTO Frequency Regulation Market Clearing Formulations and the Effect of Opportunity Costs on Electricity Prices.” arXiv:2001.11134 [esss.SY] (Jan 2020).

[C2] **Brooks, A.E.**, Lesieutre, B.C. “A Locational Price for Power Injection Fluctuations of Variable Generation and Load.” 10th IREP Bulk Power Systems Dynamics and Control Symposium, Espinho, Portugal, August 27-Sep 1, 2017.

[C3] Sehloff, D., Manur, A., **Brooks, A.E.**, Venkataramanan, G., Lesieutre, B. “Solar Forecast Uncertainty Pricing and Optimal Scheduling with the Simple Electric Utility Platform (SUEP).” MCET Renewable Energy and Sustainable Environment Conference, Pollachi, India, July 20-22, 2017.

[C4] **Brooks, A.E.**, Manur, A., Venkataramanan, G. “Energy Modeling of Aggregated Community Scale Residential Microgrids.” IEEE International Conference on Sustainable Green Buildings and Communities, Chennai, India, December 18-20, 2016.

[C5] Lai, T., **Brooks, A.E.**, Potter, B.G., Simmons-Potter, K. “Environmental Aging in Polycrystalline-Si Photovoltaic Modules: Comparison of Chamber-Based Accelerated Degradation Studies with Field-Test Data.” SPIE Optics + Photonics for Sustainable Energy, San Diego, CA, August 9-13, 2015.

[C6] **Brooks, A.E.**, Cormode, D., Cronin, A.D., Kam-Lum, E., “PV System Power Loss and Module Damage due to Partial Shade and Bypass Diode Removal Depends on Cell Behavior in Reverse Bias.” 42nd IEEE Photovoltaic Specialists Conference, New Orleans, LA, June 14-19, 2015.

[C7] Della-Giustina, D., **Brooks, A.E.**, St. Germaine, M., Patterson, S., Cronin, A. “Characterization and Use of a Sinton FMT-350 Flash Tester at the Tucson Electric Power Solar Test Yard.” OSE Optics for Solar Energy Conference, Tucson, Arizona, Nov 3-7, 2013.

[C8] Cormode, D., Cronin, A.D., Richardson, W., Lorenzo, A.T., **Brooks, A.E.**, DellaGiustina, D.N. “Comparing Ramp Rates for Large and Small PV Systems, and Selection of Batteries for Ramp Rate Control.” 39th IEEE Photovoltaic Specialists Conference, Tampa Bay, Florida, June 16-21, 2013.

[C9] **Brooks, A.E.**, DellaGiustina, D.N., Patterson, S.M., Cronin A.D. “The Consequence of Soiling on PV System Performance in Arizona; Comparing Three Study Methods.” 39th IEEE Photovoltaic Specialists Conference, Tampa Bay, Florida, June 16-21, 2013.

[C10] **Brooks, A.E.**¸Fijal, J. St. Germaine, M.T., Orsburn, S., Greenberg, J., Lonij, V.P.A., Della-Giustina, D.N., Cronin, A.D. “Conversion Efficiencies of Six Grid-Tied Inverters at the Tucson Electric Power Solar Test Yard.” 39th IEEE Photovoltaic Specialists Conference, Tampa Bay, Florida, June 16-21, 2013.

[C11] Russo, J.M., Zhang, D., Vorndran, S., Gordon, M., Castillo, J., **Brooks, A.E.**, Lonij, V., Cronin, A., Kostuk, R. “Method for site characterization of anisotropic diffuse illumination of photovoltaic systems.” SPIE Reliability of Photovoltaic Cells, Modules, Components, and Systems V Proceedings, San Diego, California, Aug 12, 2012.

[C12] **Brooks, A.E.**, Allen, N., Lonij, V.P.A., Cronin, A.D. “Evaluation of Four Geomembrane-Mounted PV Systems for Land Reclamation in Southern Arizona.” 38th IEEE Photovoltaic Specialists Conference, Austin, Texas, June 3-8, 2012.

[C13] Kopp, E.S., Lonij, V.P.A, **Brooks, A.E.**, Hidalgo-Gonzalez, P.L., Cronin, A.D. “I-V Curves and Visual Inspection of 250 PV Modules Deployed over 2 Years in Tucson.” 38th IEEE Photovoltaic Specialists Conference, Austin, Texas, June 3-8, 2012.

[C14] Hidalgo-Gonzalez, P.L., **Brooks, A.E.**, Kopp, E.S., Lonij, V.P.A., Cronin, A.D. “String-Level (kW-Scale) I-V Curves from Different Module Types under Partial Shade.” 38th IEEE Photovoltaic Specialists Conference, Austin, Texas, June 3-8, 2012.

[C15] Lonij, V.P.A., **Brooks, A.E.**, Koch, K., Cronin, A.D. “Analysis of 80 Rooftop PV Systems in the Tucson, AZ Area.” 38th IEEE Photovoltaic Specialists Conference, Austin, Texas, June 3-8, 2012.

[C16] Russo, J.M., Zhang, D., Gordon, M., Vorndran, S., Castillo, J.E., **Brooks, A.E.**, Lonij, V.P.A., Cronin, A.D., Kostuk, R.K. “Characterization of Diffuse Anistropic Illumination Effects to the Output of Bifacial and Holographic Planar Concentrating Photovoltaic Panel Configurations.” 38th IEEE Photovoltaic Specialists Conference, Austin, Texas, June 3-8, 2012.

[C17] **Brooks, A.E.**, Kostuk, R.K., Castillo, J.E., Lonij, V.P.A., Russo, J.M., Zhang, D., Vorndran, S., Cronin, A.D. "One Year of Field Studies of Holographic Planar Concentrators at the Tucson Electric Power Solar Test Yard." ASES World Renewable Energy Forum, Denver, Colorado, May 13-17, 2012.

[C18] Lonij, V.P.A., **Brooks, A.E.**, Greenberg, J., Orsburn, S., Cronin, A.D., Torres, G. "Field Performance Measurements of New and Traditional PV Technologies." ASES World Renewable Energy Forum, Denver, Colorado, May 13-17, 2012.

[C19] Cronin, A.D., Castillo, J.E., Hauser, P., Rosenberg, G.A., Kumar, R., Kostuk, R.K., Zhang, D., Russo, J.M., Vorndran, S., Lonij, V.P.A., **Brooks, A.E**. "Holographic CPV Field Tests at the Tucson Electric Power Solar Test Yard." 37th IEEE Photovoltaic Specialists Conference, Seattle, Washington, June 19-24, 2011.

[C20] Cronin, A.D., **Brooks, A.E.**, Cormode, D., Hardesty, G., Lonij, V.P.A. "Performance Reviews from the Tucson Electric Power Solar Test Yard." 37th IEEE Photovoltaic Specialists Conference, Seattle, Washington, June 19–24, 2011.

[C21] Ghosal, K., Burroughs, S., Lily, D., Gabriel, J., Cronin, A.D., **Brooks, A.E.**, Lonji, V.P.A., Krause, M.F., Lindsey, C., Metzger, S. "On-sun Performance of a Novel Microcell-Based HCPV System Based in Tucson: Comparison with Conventional Systems." 37th IEEE Photovoltaic Specialists Conference, Seattle, Washington, June 19–24, 2011.

Significant Media Contributions

Additional recorded presentations and interviews available online

[M1] Tsirkin, J., "Inside the battle for clean energy on Native American land," NBC News, online (Jul 2026)

[M2] "Transmission Planning 101: Understanding the Grid," U.S. Department of Energy Grid Deployment Office, online (Aug 2024)

[M3] Brouns, Z., McCaskey, K., "Dr. Adria Brooks, Grid Deployment Office, Transmission Champion," Federation of American Scientists, (Jan 2024)

[M4] Howland, E., "Pending Senate bill would require 30% transfer capacity between regions: Hickenlooper aide," Utility Dive (Dec 2023)

[M5] Popovich, N., Plumer, B., "Why the U.S. Electric Grid Isn't Ready for the Energy Transition," New York Times (Jun 2023)

[M6] Howland, E., "DOE finds 'pressing' transmission needs, with interregional links offering the biggest benefits," Utility Dive (Mar 2023)

[M7] Howland, E., "FERC urged to set interregional transfer capacity requirements to boost reliability, lower costs," Utility Dive (Dec 2022)

Awards & Distinctions

Excellence Award ***For technical contributions in transmission value assessments and leadership on seminal national transmission studies*** Energy Systems Integration Group, Tucson AZ (Mar 2026)

Early Career Award University of Wisconsin, College of Engineering, Madison WI (Nov 2025)

Secretary of Energy Achievement Award (2) Department of Energy, Washington DC (Jan 2023 and Jan 2024)

Science and Technology Policy Fellowship Program American Association for the Advancement of Science, Washington DC (Aug 2020 – Oct 2021)

Grainger Power Engineering Award University of Wisconsin-Madison, Madison WI (April 2019)

Graduate Research Fellowship Program National Science Foundation, Washington DC (March 2016)

Wisconsin Distinguished Graduate Fellowship University of Wisconsin-Madison, Madison WI (August 2015)

Bill of Rights Award American Civil Liberties Union of Tennessee, Nashville TN (December 2008)

NASA Motivating Undergraduates in Science & Technology Award University of Arizona, Tucson AZ (May 2008)

First Place in Undergraduate Engineering Research University of Arizona, Tucson AZ (April 2008)

Academic Distinction Award University of Arizona, Tucson AZ (October 2006)

Arizona Excellence Scholarship for Undergraduate Studies University of Arizona, Tucson AZ (August 2005)

Professional Activities

Memberships

- IEEE Power and Energy Society & Women in Engineering (2019 – 2022, 2025-present)
- New England Women in Energy and the Environment (2022 – 2024)
- American Association of the Advancement of Science (2019 – 2023)
- National Science Policy Network (2019 – 2020)

Activities

- Evergreen Data Visualization Academy (2020 - 2021)
- Invited editor, IEEE and Elsevier on papers related to transmission planning and system modeling (2018 - 2022)
- Diversity Equity and Inclusion in the Workplace Certificate, University of South Florida Muma College of Business (2021)
- Data Visualization for Storytellers short course, University of California Berkeley (2021)
- Influencing Virtually short course, Georgetown University McCourt School of Public Policy (2021)
- Mentor with TryEngineering (2019 – 2020)
- Televised interview with Arizona Public Media for “Metro Week: Solar Energy Advances in the Southwest” (June 2015)
- MSHA Part 46 Surface Miner Certification (2011 – 2013)
- Solar Energy International Solar Electric Design and Installation Workshop, Guemes Island WA (2011)

Grants & Contracts

- Photovoltaic Crystalline-Silicon System Field Performance Comparative Study, SunPower Corporation, $150,000 (2013)
- FRV PV Heat-Islanding Monitoring Project of Interest, Fotowatio Renewable Ventures, $100,000 (2012)
- Photovoltaic System Field Performance Comparative Study, Rosemont Copper, $145,000 (2011)

Leadership and Community Service Activities

- Southern Arizona Rescue Association, Mountain Wilderness Rescuer (2024 - present)
- Teaching Assistants’ Association, South Central Federal of Labor Delegate (2018 – 2020), Co-President (2016 – 2018), departmental steward (2015 – 2020)
- *The WEMPECKER* (departmental annual satirical paper), Lead Editor (2016 – 2018)
- Mountain Vista Unitarian Universalist religious education Chairperson (2014) and Youth Director (2012 – 2014)
- EON Youth Lounge volunteer (2005 – 2006 and 2011 – 2014)
- Camp Wildcat Vice-Chairperson (2010 – 2011), Secretary (2007 – 2008), youth counselor (2005 – 2011)
- One-In-Teen Youth Services Board Member (2003 – 2005)

References

Ms. Gretchen Kershaw
Chief Operating Officer and Vice President of Strategy, Grid Strategies LLC, Washington DC
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Mrs. Maria Robinson
Former Director, Grid Deployment Office, Department of Energy, Washington DC
mduaime@alum.mit.edu

Dr. Elaine Hale
Researcher IV – Model Engineering, National Renewable Energy Laboratory, Golden CO
elaine.hale@nrel.gov

Dr. Bernie Lesieutre
Professor of Electrical Engineering, University of Wisconsin, Madison WI
lesieutre@engr.wisc.edu